

Week 1 Assignment: “Clean Enough to Trust?”

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Data Set Description

Breaking down the systems and finding the logic behind the data is an important prerequisite for implementing any model. The selected data set called "business_data.csv" acts as the starting point for assessing the behavior and classification of retail customers. The data set was obtained from the repository CS-540 Module One Assignment. As per the scope of this study, the credibility of the data set will be considered reliable since the data set is provided by the institution itself. This dataset consists of a total of 1,800 data entries spread across seven distinct variables. These seven variables can be classified into two broad categories according to their data type. There are four variables in total whose data entries consist of numeric values. These include CustomerID (which is a unique identification code), Age, AnnualIncome, and SpendingScore.

Identification of Data Quality Issues

Prior to mining any insights from the data set, it is imperative that the logic behind the data is audited. There are several data quality concerns that must be addressed in the file that would adversely affect analysis. Firstly, there is a high prevalence of missing data points. The Gender field has no data present in 252 data points, accounting for 14% of the data set. Additionally, the Membership field has "hidden" missing data points where blank fields have been entered. The second problem is that the data is extremely inconsistent with regard to formatting. For example, the product type is represented in the form of "COMPUTER," "electronics," "DRinkS," and "gUITARS." Third, there is some extremely weird and mathematically impossible data. For instance, the age ranges from 0 to 150, resulting in a median

age of 1 for all 1,800 observations. Moreover, there is at least one meaningless category called "dxm xp8" in the product type.

Evaluation of Overall Quality and Cleanliness

The quality of the data set is exceptionally low. As it stands now, this data set would be completely unusable in any operational or business intelligence setting. The combination of improper formatting and logical impossibilities is indicative of inadequate data validation at an earlier stage. In terms of a specific example, we can look no further than the row index at 4 to see an illustration of what I mean when talking about data that is dirty. What we see here is an observation that refers to customer number 12158261, who is a Female with an age of 126 years, which seems statistically unlikely. In addition, the membership level she has been given is simply a blank field. The purchased product type was also misspelled.

Impact on Effective Analysis

Should the above data be used as is, then the analysis outcomes would be significantly jeopardized. As seen from price action anomaly data that can generate incorrect trading signals for algorithms to swing trade on, the noise found within the above data will skew mathematical summations and logic breakdowns. For example, the inclusion of customers aged 0 and 150 makes mathematical sense for any form of demography impossible. Finding the average customer age will create a very skewed figure which will provide erroneous information that might lead to wrong marketing practices. Another issue is that grouping algorithms can be quite literal when reading the data set since the inconsistent entries in the ProductType column make "electronics" and "ELECTRONICS" appear like different types of products. This leads to a

misunderstanding of the number of sales made in each particular category. Finally, 14% lack of gender information makes up a demographic gap.

Potential Questions or Insights

After having cleansed the underlying data logic, there are various valuable uses that can be derived from this dataset in terms of formulating optimal retail strategies. Firstly, the dataset provides answers to the issue of whether AnnualIncome and SpendingScore are related; in other words, it is important to know whether high AnnualIncome predicts higher SpendingScore or the middle brackets bring more sales volume. Moreover, the company will learn about the effectiveness of its loyalty program. In particular, the company might find out whether the premium members such as Gold and Silver bring more sales in particular ProductTypes than the members of the Standard category. Lastly, after cleansing the underlying Age and Gender distribution, it will be possible to identify key demographics with high SpendingScore.

AI Usage Acknowledgement

In preparing this assignment, I used the help of Google's Gemini AI in creating a walkthrough plan in profiling the data, and detecting data quality problems from the raw data. All information provided by the AI was checked and validated from the raw business_data.csv file before being submitted.